

GNU-Plot-Lab-Notebook

Roll Number: _____

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Semester: _____

Section 1

Question 1

Problem Set 1 (xvii). Define the piecewise function

$$f(x) = \begin{cases} x^2, & x < 1, \\ x, & 1 \leq x \leq 2, \\ x^3/4, & x > 2. \end{cases}$$

Plot it for $0 \leq x \leq 3$ and $0 \leq y \leq 8$, with x -ticks every 0.5, y -ticks every 1, and the title *Polynomials of x* .

Sol: The GNUplot script and its output are shown below.

```
1  reset
2  set term svg background "white"
3  set ou "Lab-files/Output/1/Q_xvii_ou.svg"
4
5
6
7  set title "Polynomials of x"
8
9  set grid
10
11 set xzeroaxis linestyle 1 lc "black"
12 set yzeroaxis linestyle 1 lc "black"
13
14 set xrange [0:3]
15 set yrange [0:8]
16
17 set xtics 0.5
18 set ytics 1
19
20 f(x) = (x<1)?(x**2) : (x>=1 && x<=2)?(x) : (x>2)?((x**3)/4):NaN
21
22 plot f(x) lc "#CC79A7" lw 2
```

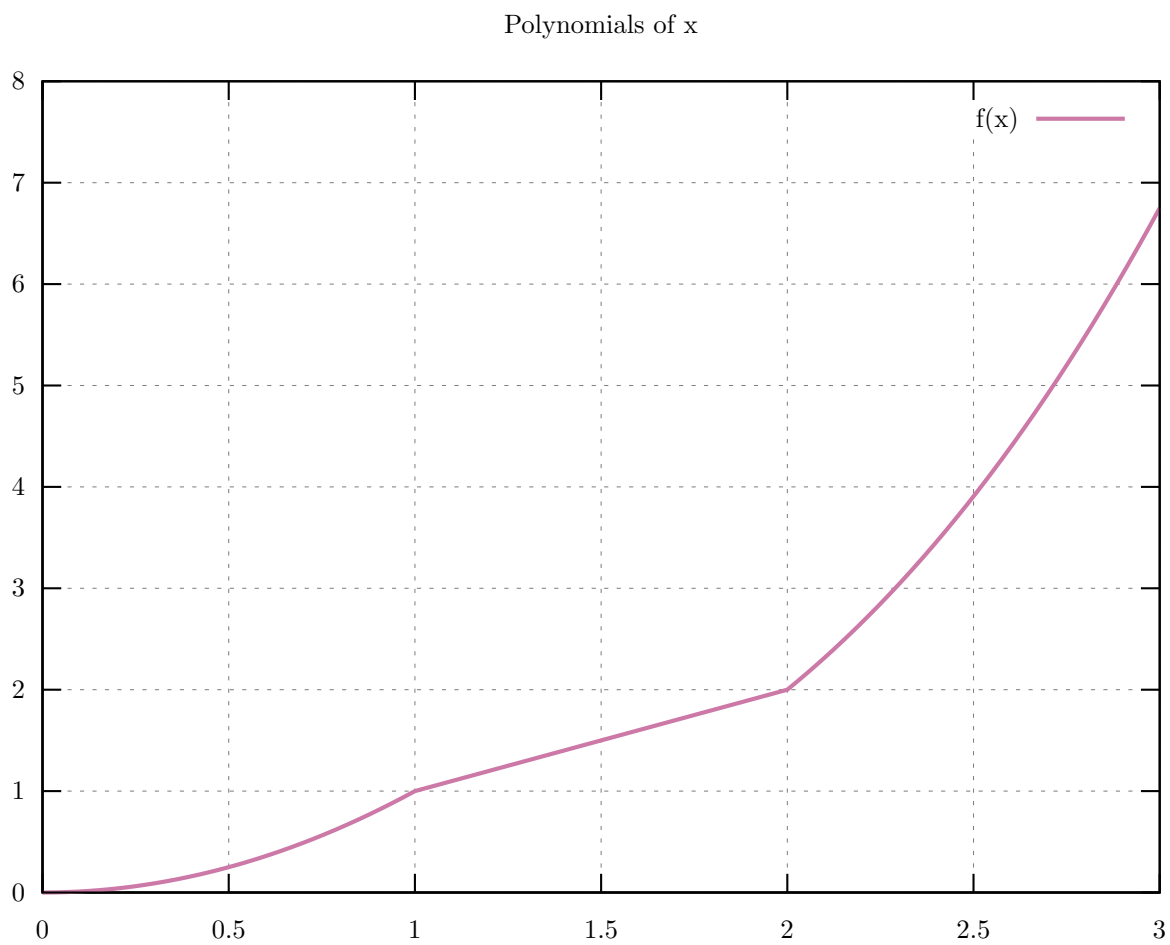


Figure 1: Output for Problem Set 1 (xvii).

Question 2

Problem Set 1 (xv). Plot $y = x \sin(3x)$ for $-3 \leq x \leq 3$ and $-4 \leq y \leq 4$. Show both axes, label them x values and y values, and set the x - and y -tick intervals to 0.5 and 1, respectively.

Sol: The GNUplot script and its output are shown below.

```
1  reset
2  set term svg background "white"
3  set ou "Lab-files/Output/1/Q_xv_ou.svg"
4
5
6
7
8
9  set grid
10
11 set xzeroaxis linestyle 1 lc "black"
12 set yzeroaxis linestyle 1 lc "black"
13
14 set xrange [-3:3]
15 set yrange [-4:4]
16
17 set xlabel "x values"
18 set ylabel "y values"
19
20 set xtics 0.5
21 set ytics 1
22
23
24 y(x)=x*sin(3*x)
25
26 plot y(x) lc "#009E73" lw 2
```

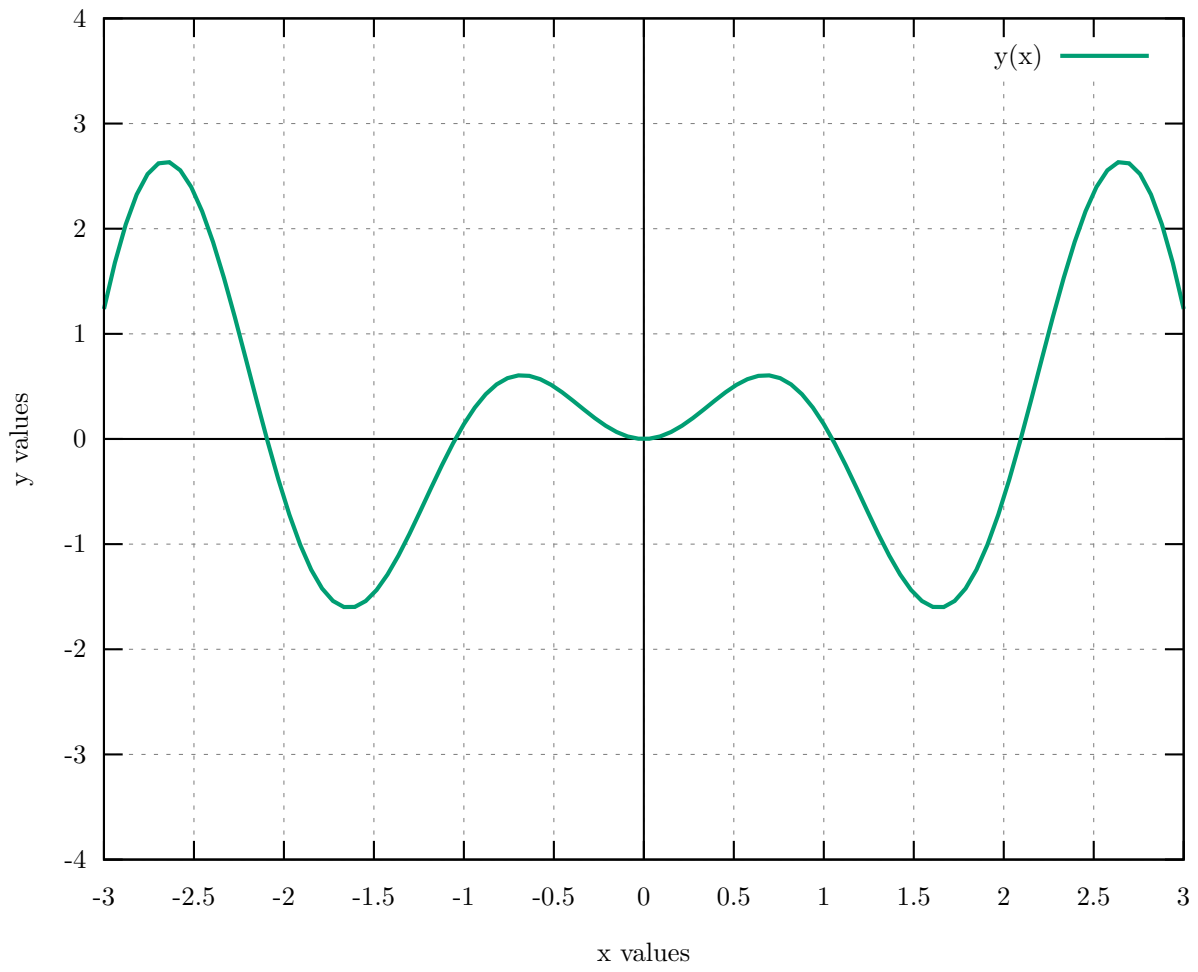


Figure 2: Output for Problem Set 1 (xv).

Question 3

Problem Set 1 (i). Plot $y = f(x) = x^5 + 2x^2 + 6$ for $-10 \leq x \leq 10$. Show both axes, label them `x values` and `new function`, and set the x -tick interval to 2.

Sol: The GNUplot script and its output are shown below.

```
1  reset
2  set terminal svg background "white"
3
4  set ou "Lab-files/Output/1/Q_i_ou.svg"
5
6  set grid
7
8  set xrange [-10:10]
9  set yrange [-110000:110000]
10
11 set xzeroaxis linestyle 1 lc "black"
12 set yzeroaxis linestyle 1 lc "black"
13
14 set xtics 2
15
16
17 set xlabel "x values"
18 set ylabel "new function"
19
20
21 f(x)=x**5 + 2*x**2 + 6
22
23 plot f(x) lc "#0072B2" lw 2
```

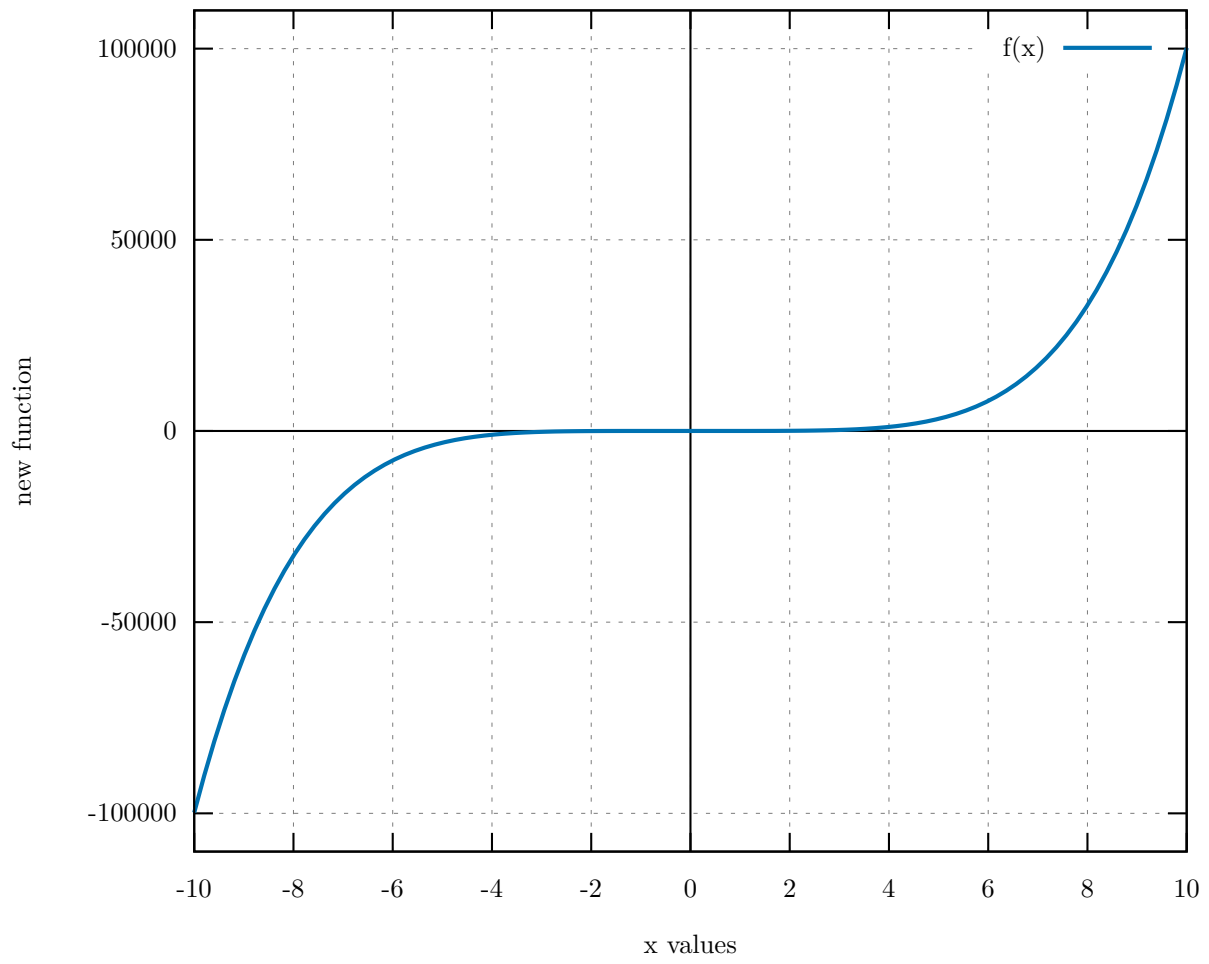


Figure 3: Output for Problem Set 1 (i).

Question 4

Problem Set 1 (xi). Plot $y = \ln(x^2)$ for $-2 \leq x \leq 2$ and $-3 \leq y \leq 3$. Show both axes, label them x values and y values, and set the x - and y -tick intervals to 0.5 and 1, respectively.

Sol: The GNUplot script and its output are shown below.

```
1  reset
2  set terminal svg background "white"
3  set ou "Lab-files/Output/1/Q_xi_ou.svg"
4
5  set grid
6
7  set xrange [-2:2]
8  set yrange [-3:3]
9
10 set xtics 0.5
11 set ytics 1
12
13
14 set xlabel "x values"
15 set ylabel "y values"
16
17 set xzeroaxis linestyle 1 lc "black"
18 set yzeroaxis linestyle 1 lc "black"
19
20 y(x)=log(x**2)
21
22 plot y(x) lc "#D55E00" lw 2
```

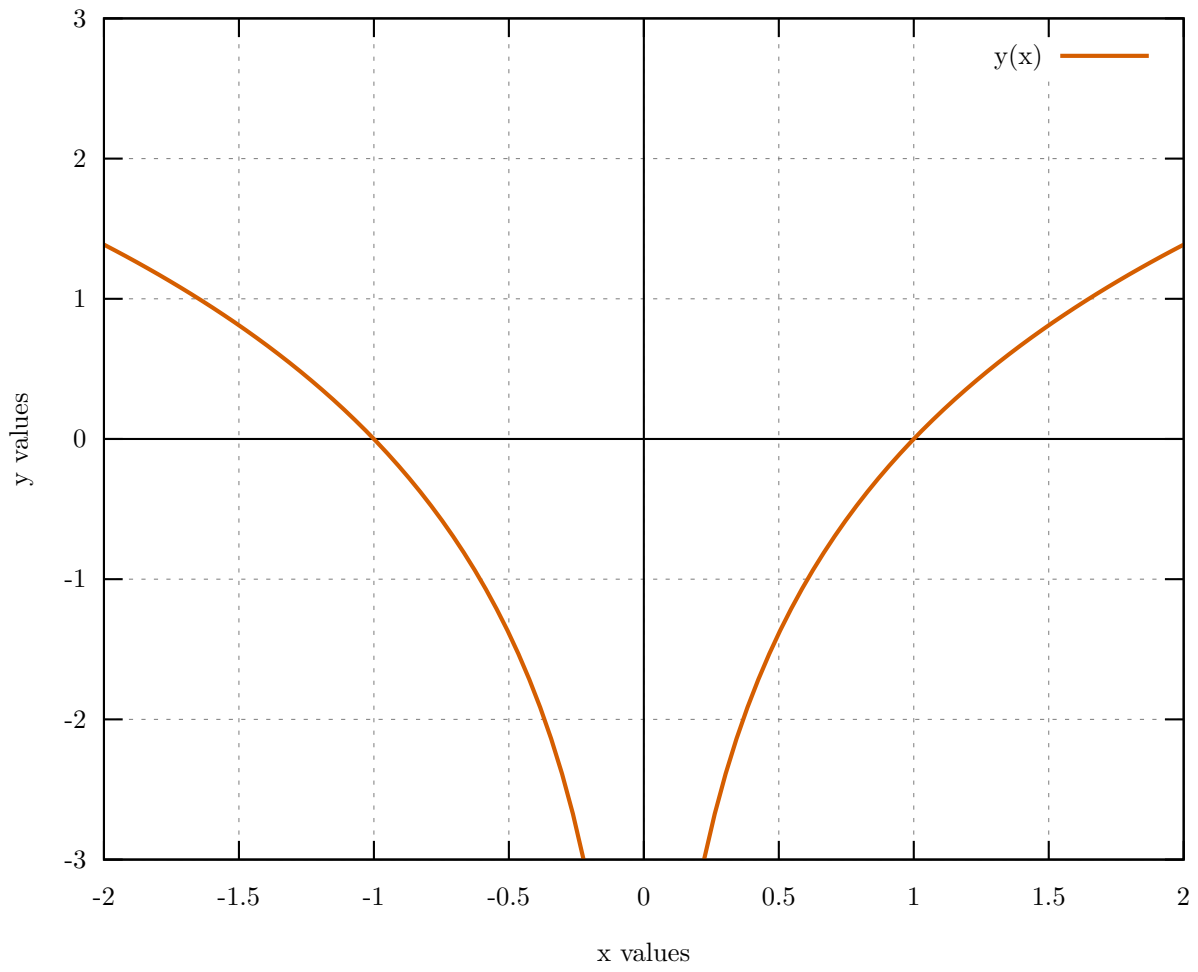


Figure 4: Output for Problem Set 1 (xi).

Question 5

Problem Set 1 (vi). Plot $f(x) = 2x^3 + 9x - 11$ for $-15 \leq x \leq 10$ and $-7000 \leq y \leq 2200$. Show the y -axis, label the axes `x value` and `new function`, and set the x -tick interval to 2.5.

Sol: The GNUplot script and its output are shown below.

```
1  reset
2  set terminal svg background "white"
3
4  set ou "Lab-files/Output/1/Q_vi_ou.svg"
5
6
7  f(x)=2*x**3 + 9*x - 11
8
9  set xrange [-15:10]
10 set yrange [-7000:2200]
11
12 set grid
13
14 set xzeroaxis linestyle 1 lc "black"
15 set yzeroaxis linestyle 1 lc "black"
16
17 set xtics 2.5
18
19 set xlabel "x value"
20 set ylabel "new function"
21
22
23 plot f(x) lw 2 lc "#CC79A7"
```

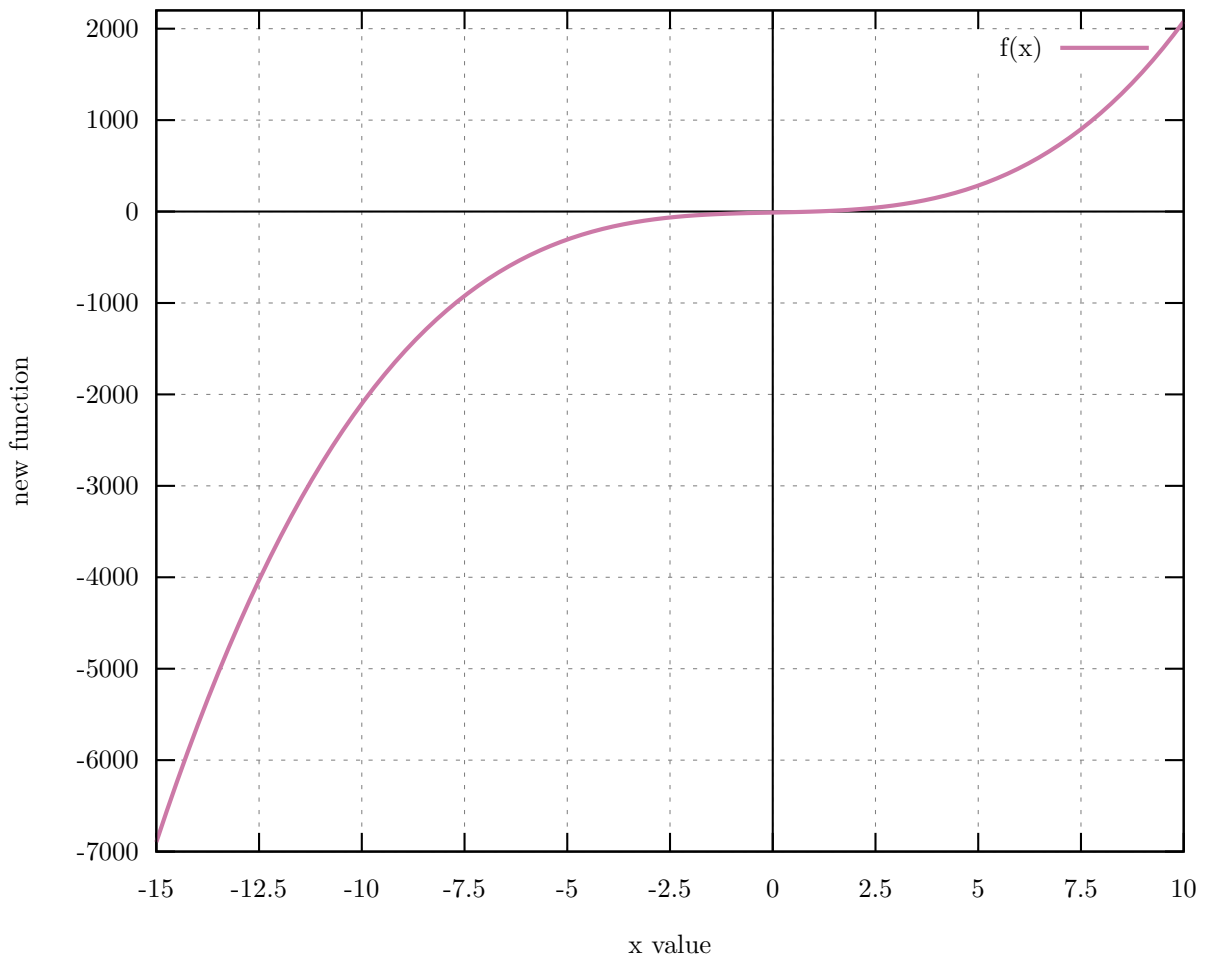


Figure 5: Output for Problem Set 1 (vi).

Question 6

Problem Set 1 (v). Plot the piecewise functions

$$f_1(x) = x \ (x > 4), \quad f_2(x) = -x^2 \ (-4 < x < 4), \quad f_3(x) = -x \ (x < -4)$$

for $-10 \leq x \leq 10$ and $-4 \leq y \leq 10$. Set the x -tick interval to 1 and label the axes `x values` and `piecewise continuous functions`.

Sol: The GNUplot script and its output are shown below.

```
1  reset
2  set terminal svg background "white"
3  set ou "Lab-files/Output/1/Q_v_ou.svg"
4
5  set grid
6
7  set xrange [-10:10]
8  set yrange [-4:10]
9
10 set xtics 1
11
12 set xlabel "x values"
13 set ylabel "piecewise continuous functions"
14
15 set xzeroaxis linestyle 1 lc "black"
16 set yzeroaxis linestyle 1 lc "black"
17
18
19 f3(x)=(x<-4)?(-x):NaN #Not a Number
20 f2(x)=(x>-4 && x<4)?(-x**2):NaN
21 f1(x)=(x>4)?(x):NaN
22
23
24 plot f1(x) lc "#0072B2" lw 2 ,f2(x) lc "#A65E00" lw 2 ,f3(x) lc "#CC79A7" lw 2
```

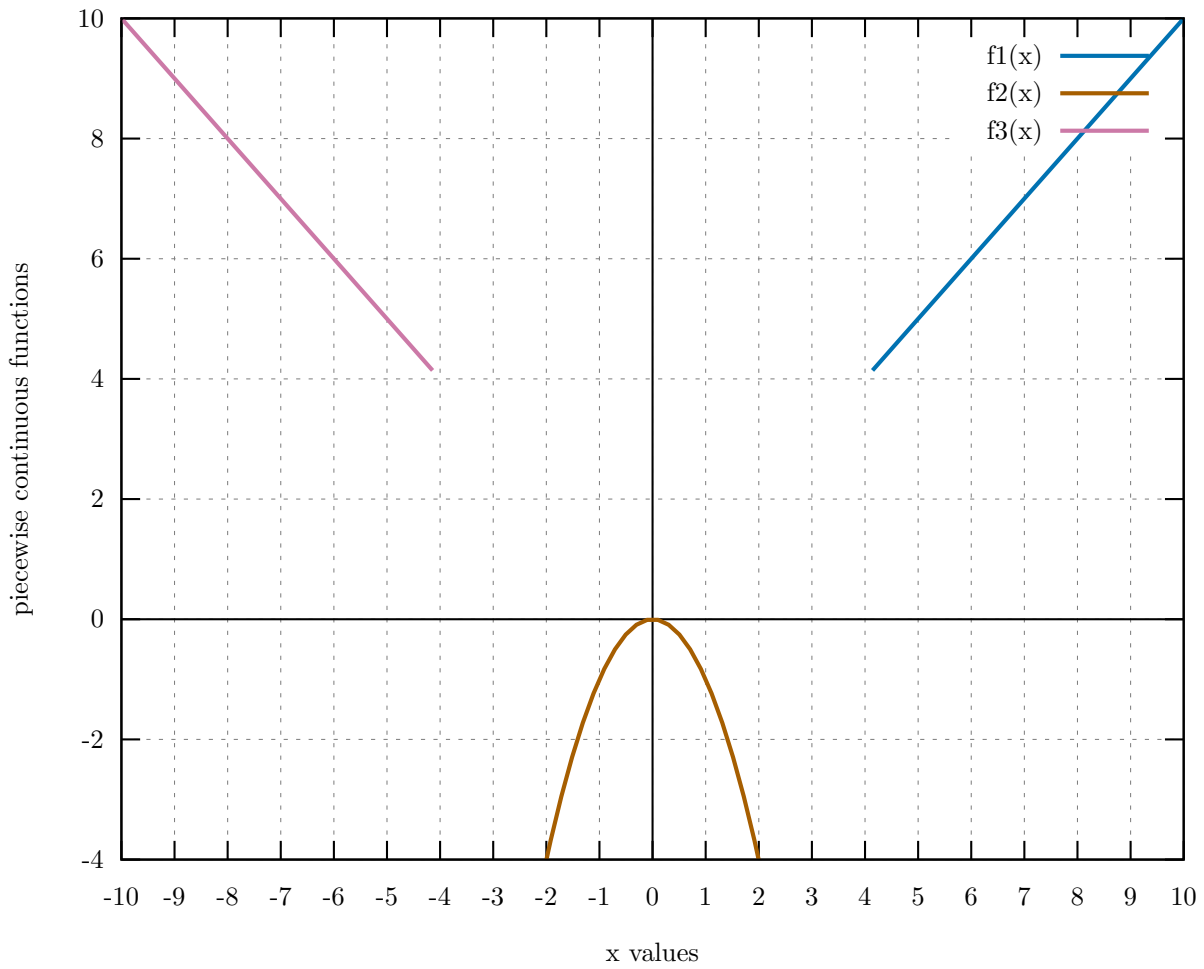


Figure 6: Output for Problem Set 1 (v).

Question 7

Problem Set 1 (iv). Plot the piecewise function

$$f(x) = \begin{cases} x, & -2 \leq x \leq 2, \\ 0.5, & \text{otherwise} \end{cases}$$

for $-10 \leq x \leq 10$ and $-5 \leq y \leq 5$. Set the x - and y -tick intervals to 2 and 1, respectively, and label the axes `x value` and `jerk function`.

Sol: The GNUplot script and its output are shown below.

```
1  reset
2  set terminal svg background "white"
3  set ou "Lab-files/Output/1/Q_iv_ou.svg"
4
5  set grid
6
7  set xtics 2
8  set ytics 1
9
10 set xzeroaxis linestyle 3 lc "black"
11 set yzeroaxis linestyle 3 lc "black"
12
13 set xrange [-10:10]
14 set yrange [-5:5]
15
16 set xlabel "x value"
17 set ylabel "jerk function"
18
19 f(x)=(x>=-2 && x<=2)?(x):0.5
20
21 plot f(x) lc "#009E73" lw 2
```

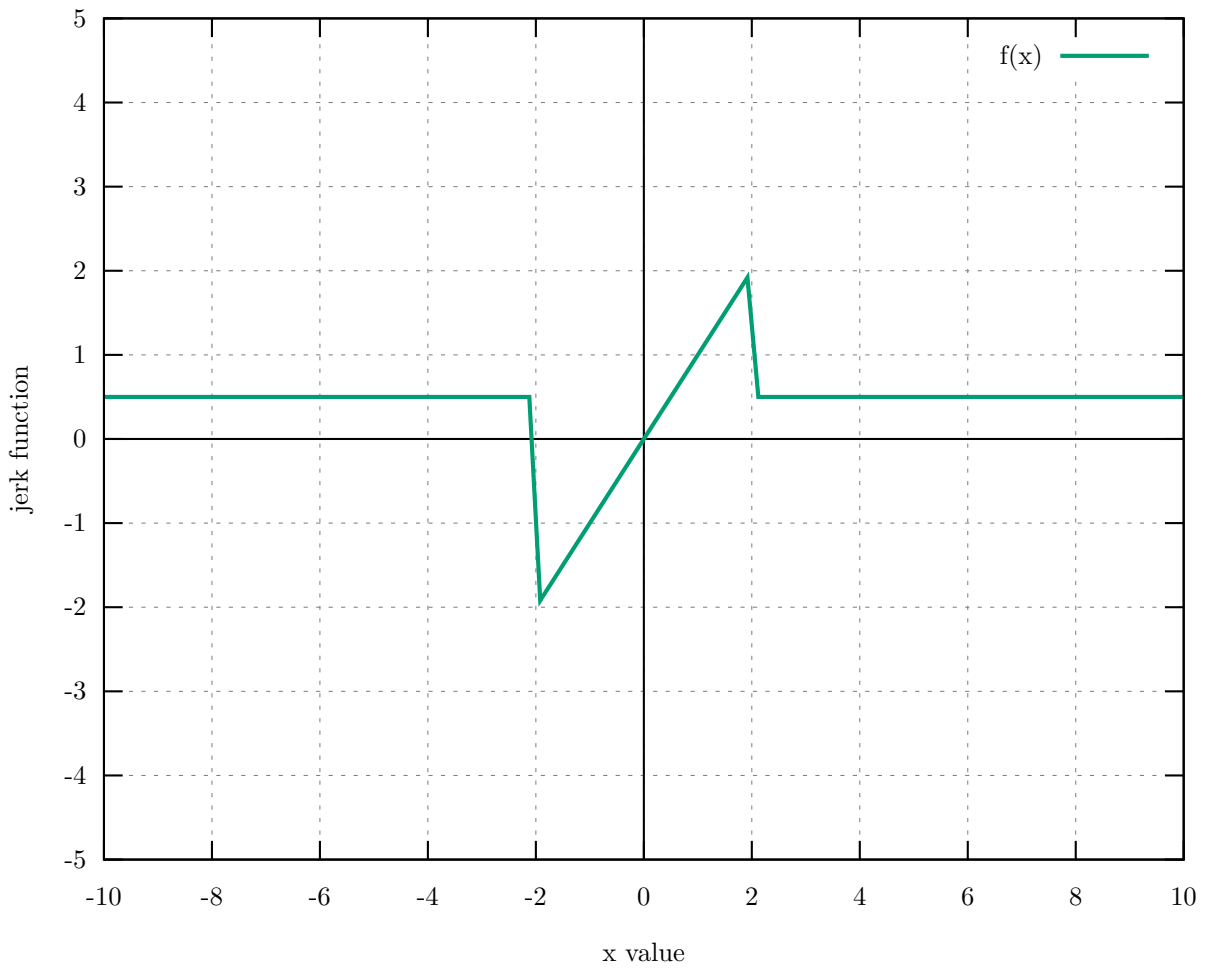


Figure 7: Output for Problem Set 1 (iv).

Section 2

Question 1

Problem Set 2 (ii). Fit the data with $y = mx + c$ over $-7 \leq x \leq 7$. Plot the points and fitted line, show the y -axis, and choose the y -range so c can be read from that axis. Report m and c .

Table 1: Data for Problem Set 2 (ii).

x	y
-5.1	21.8
-3.4	16.12
-0.5	7.05
1.57	0.42
6.2	-14.2

Data file contents:

```
-5.1 21.8  
-3.4 16.12  
-0.5 7.05  
1.57 0.42  
6.2 -14.2
```

Sol: The fitted plot and the parameter estimates with their standard errors are shown below.

```
1  reset  
2  set terminal svg background "white"  
3  set ou "Lab-files/Output/2/Q_2_ii_ou.svg"  
4  
5  
6  set fit logfile "Lab-files/GP-Files/2/Fit-log/fit_ii.log"  
7  
8  
9  set grid  
10 set xzeroaxis linestyle 0 lc "black"  
11 set yzeroaxis linestyle 0 lc "black"  
12  
13 set xrange [-7:7]  
14 #Fitting_curve  
15 y(x)=m * x + c  
16  
17 #Initial guess  
18 m=-1  
19 c=5  
20 #Curve fitting  
21 fit y(x) "Lab-files/Data/2/Q_ii.txt" using 1:2 via m,c  
22 #Plotting  
23 plot "Lab-files/Data/2/Q_ii.txt" using 1:2 with p ps 1.5 lc "#009E73" title "Data  
    points", \  
24     y(x) lw 2 lc "#CC79A7" title "Fitted line"
```

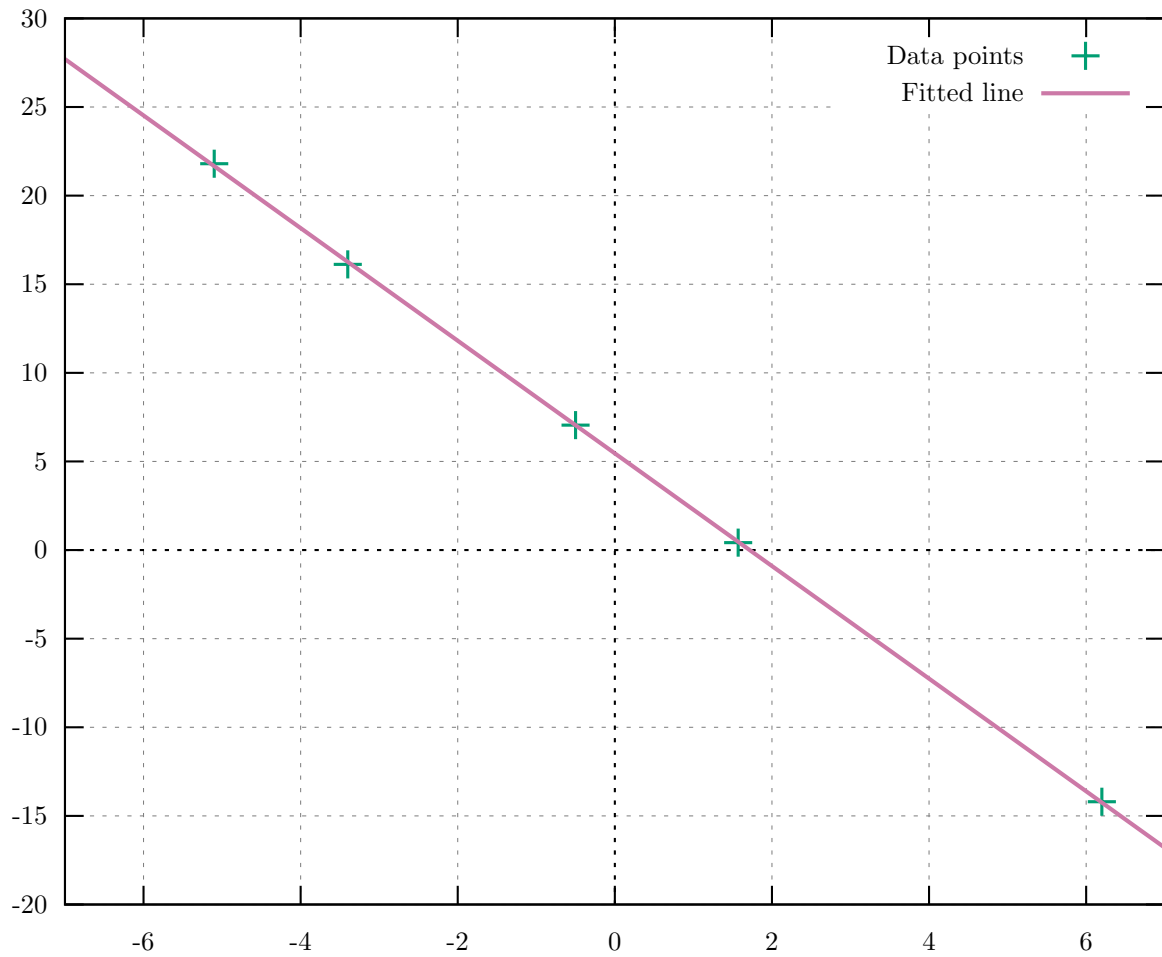


Figure 8: Linear fit for Problem Set 2 (ii).

Fit log (parameter estimates and standard errors):

Final set of parameters		Asymptotic Standard Error	
=====		=====	
m	= -3.17793	+/- 0.01351	(0.4251%)
c	= 5.45623	+/- 0.05361	(0.9825%)

Question 2

Problem Set 2 (iii). Fit the data with $y = mx + c$ over $0 \leq x \leq 9$. Plot the points and fitted line, show the y -axis, and choose the y -range so c can be read from that axis. Report m and c .

Table 2: Data for Problem Set 2 (iii).

x	y
2.1	-2.5
4.6	7.7
5.7	12.17
6.6	15.9
8.2	22.4

Data file contents:

```
2.1 -2.5
4.6 7.7
5.7 12.17
6.6 15.9
8.2 22.4
```

Sol: The fitted plot and the parameter estimates with their standard errors are shown below.

```
1  reset
2  set terminal svg background "white"
3  set ou "Lab-files/Output/2/Q_2_iii_ou.svg"
4
5
6  set fit logfile "Lab-files/GP-Files/2/Fit-log/fit_iii.log"
7
8  set grid
9  set xzeroaxis linestyle 0 lc "black"
10 set yzeroaxis linestyle 0 lc "black"
11 set xrange [0:9]
12
13 #Fitting Curve
14
15 y(x)= m * x + c
16
17 #Initial Guess
18 c=-10
19 m=2
20 #Curve Fitting
21 fit y(x) "Lab-files/Data/2/Q_iii.txt" using 1:2 via m,c
22 #Plotting
23
24 plot "Lab-files/Data/2/Q_iii.txt" using 1:2 with p lc "#0072B2" ps 1.5 title "Data
    points", \
25      y(x) lw 2 lc "#D55E00" title "Fitted line"
```

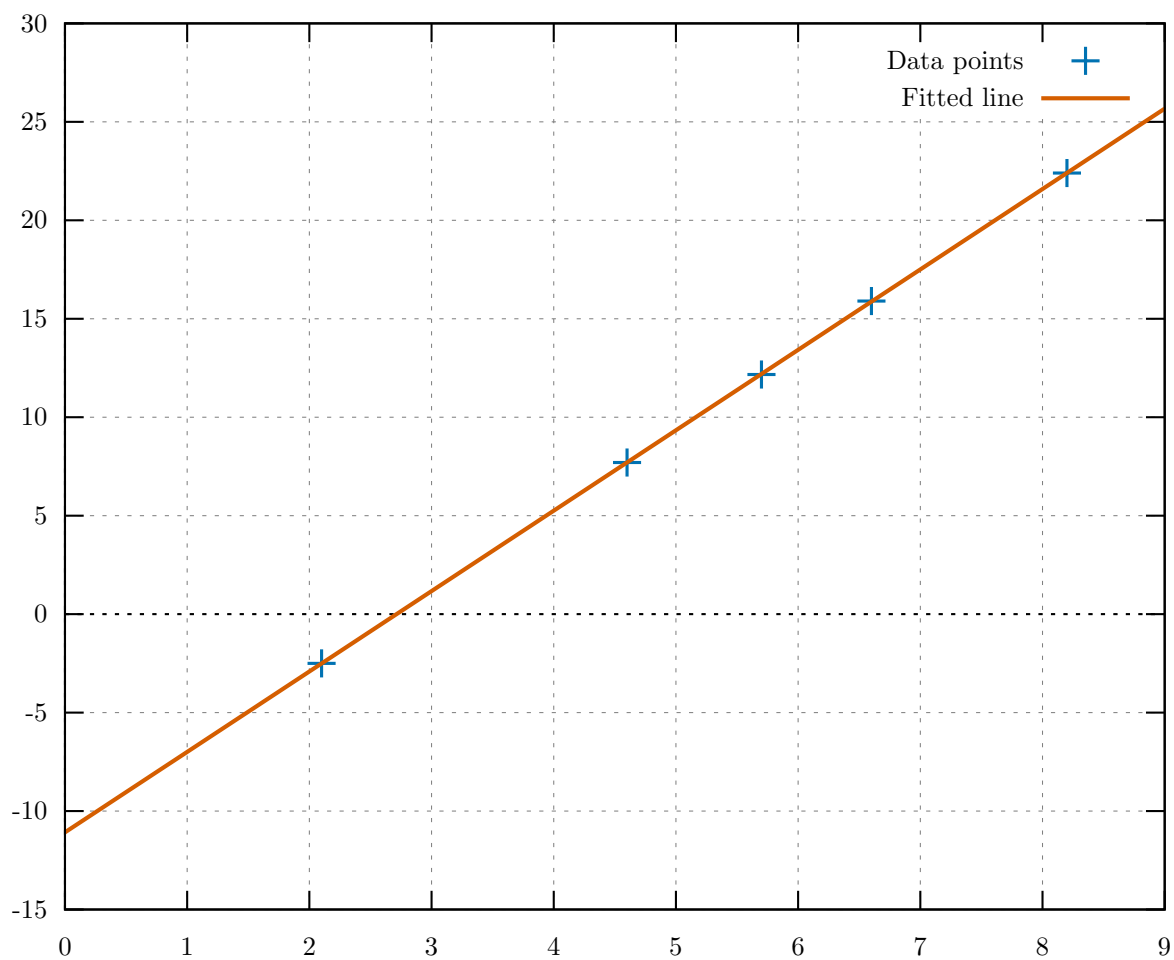


Figure 9: Linear fit for Problem Set 2 (iii).

Fit log (parameter estimates and standard errors):

Final set of parameters		Asymptotic Standard Error	
=====		=====	
m	= 4.08358	+/- 0.005003	(0.1225%)
c	= -11.0807	+/- 0.02907	(0.2624%)

Question 3

Problem Set 2 (iv). Fit the data with $f(x) = mx + c$ over $-3 \leq x \leq 4$. Plot the points and fitted line, show both axes, and choose the y -range so c can be read from the y -axis. Report m and c .

Table 3: Data for Problem Set 2 (iv).

x	y
-2.0	-4.15
-1.3	-2.40
1.8	6.60
2.5	8.98
3.6	11.94

Data file contents:

```
-2.0 -4.15  
-1.3 -2.40  
1.8 6.60  
2.5 8.98  
3.6 11.94
```

Sol: The fitted plot and the parameter estimates with their standard errors are shown below.

```
1  reset  
2  set terminal svg background "white"  
3  set ou "Lab-files/Output/2/Q_2_iv_ou.svg"  
4  
5  
6  set fit logfile "Lab-files/GP-Files/2/Fit-log/fit_iv.log"  
7  
8  set grid  
9  set xzeroaxis linestyle 0 lc "black"  
10 set yzeroaxis linestyle 0 lc "black"  
11  
12 set xrange [-3:4]  
13  
14 y(x)=m*x+c  
15 c=0.1  
16 m=1  
17  
18 fit y(x) "Lab-files/Data/2/Q_iv.txt" using 1:2 via m,c  
19  
20 plot "Lab-files/Data/2/Q_iv.txt" using 1:2 ps 1.5 lc "#CC79A7" title "Data points"  
21 , \  
    y(x) lw 2 lc "#009E73" title "Fitted line"
```

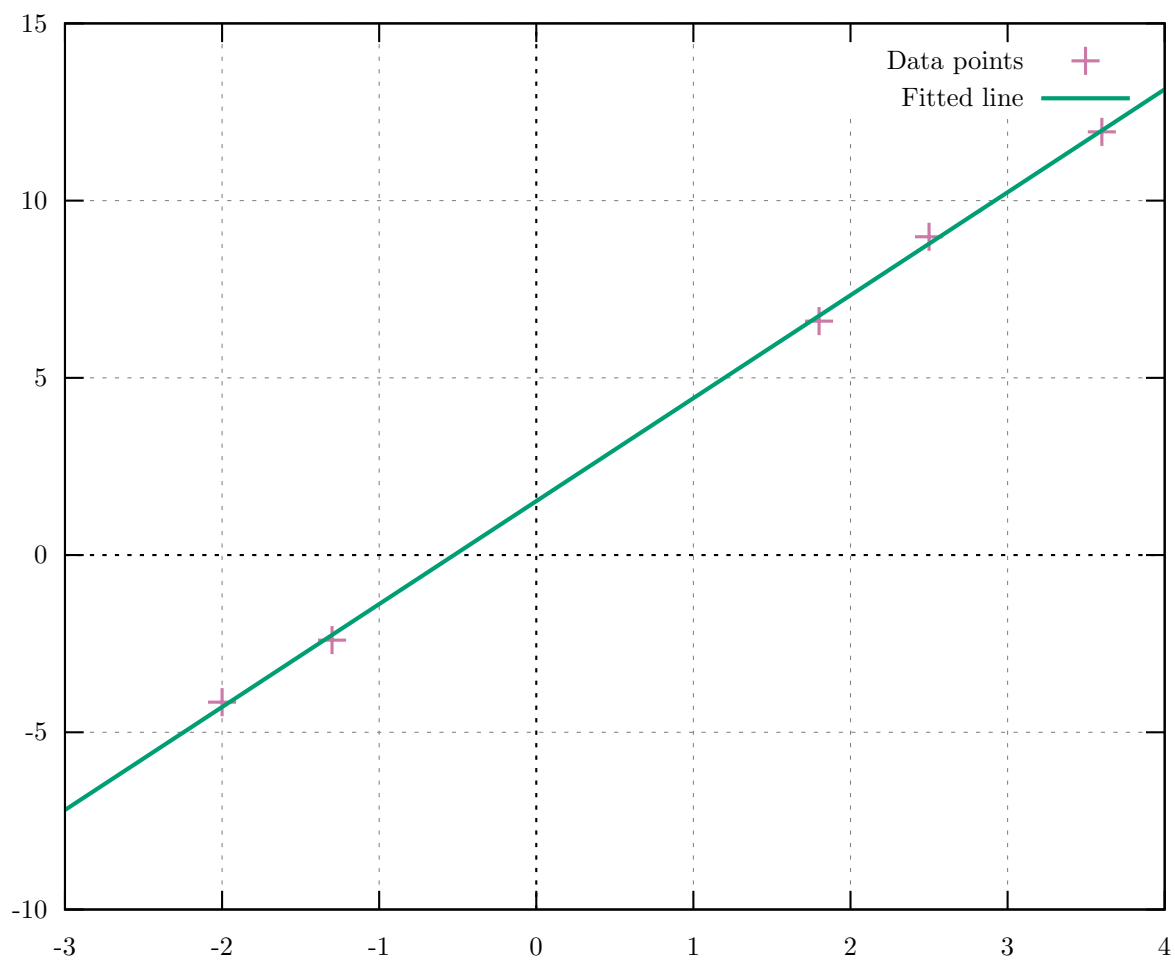


Figure 10: Linear fit for Problem Set 2 (iv).

Fit log (parameter estimates and standard errors):

Final set of parameters		Asymptotic Standard Error	
=====		=====	
m	= 2.90453	+/- 0.03786	(1.304%)
c	= 1.52183	+/- 0.08983	(5.902%)

Question 4

Problem Set 2 (v). Fit the data with $f(x) = ax^b$ over $0 \leq x \leq 3$ and $0 \leq y \leq 35$. Plot the points and fitted curve, use the title *Power series fit*, and report a and b .

Table 4: Data for Problem Set 2 (v).

x	y
0.5	0.21
0.83	1.05
1.16	2.40
1.82	9.01
2.81	30.75

Data file contents:

```
0.5 0.21
0.83 1.05
1.16 2.40
1.82 9.01
2.81 30.75
```

Sol: The fitted plot and the parameter estimates with their standard errors are shown below.

```
1  reset
2  set terminal svg background "white"
3  set ou "Lab-files/Output/2/Q_2_v_ou.svg"
4
5
6  set fit logfile "Lab-files/GP-Files/2/Fit-log/fit_v.log"
7
8  set title "Power series fit"
9  set grid
10 set xzeroaxis linestyle 0 lc "black"
11 set yzeroaxis linestyle 0 lc "black"
12
13 set xrange [0:3]
14 set yrange [0:35]
15
16 y(x)=a*(x**b)
17
18 a=1
19 b=1
20
21 fit y(x) "Lab-files/Data/2/Q_v.txt" using 1:2 via a,b
22
23 plot "Lab-files/Data/2/Q_v.txt" using 1:2 with p ps 1.5 lc "#D55E00" title "Data
    points", \
24     y(x) lw 2 lc "#0072B2" title "Fitted curve"
```

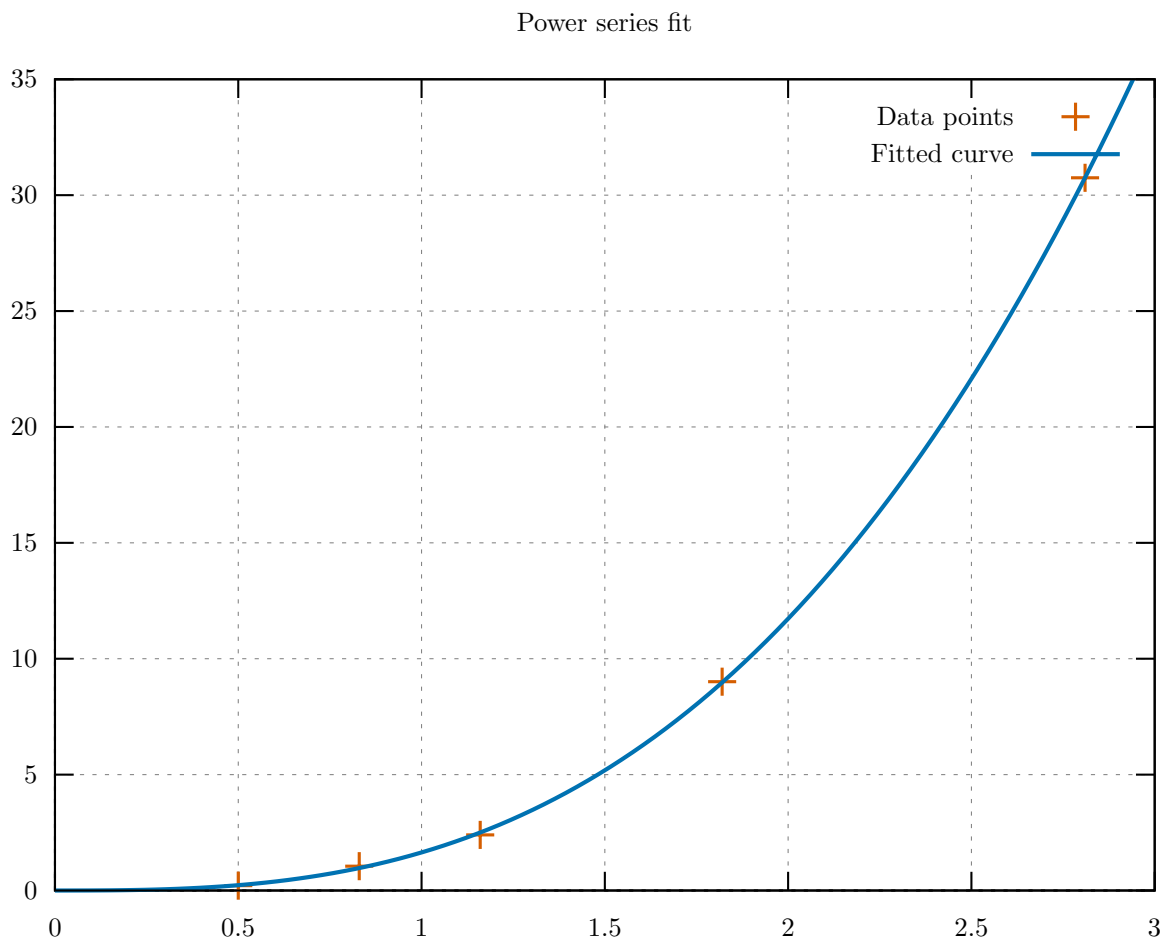


Figure 11: Power-series fit for Problem Set 2 (v).

Fit log (parameter estimates and standard errors):

Final set of parameters		Asymptotic Standard Error	
=====		=====	
a	= 1.64269	+/- 0.02914	(1.774%)
b	= 2.8356	+/- 0.01767	(0.6232%)

Question 5

Problem Set 2 (vi). Fit the data with $f(x) = mx + c$ over $-3 \leq x \leq 3$. Plot the points and fitted line, show both axes, and choose the y -range so c can be read from the y -axis. Report m and c .

Table 5: Data for Problem Set 2 (vi).

x	y
-2.9	-5.96
-1.4	-3.81
0.8	-1.10
1.7	0.21
2.5	1.30

Data file contents:

```
-2.9 -5.96
-1.4 -3.81
0.8 -1.10
1.7 0.21
2.5 1.30
```

Sol: The fitted plot and the parameter estimates with their standard errors are shown below.

```
1  reset
2  set terminal svg background "white"
3  set ou "Lab-files/Output/2/Q_2_vi_ou.svg"
4
5
6  set fit logfile "Lab-files/GP-Files/2/Fit-log/fit_vi.log"
7
8  set grid
9  set xzeroaxis linestyle 0 lc "black"
10 set yzeroaxis linestyle 0 lc "black"
11
12 set xrange [-3:3]
13
14 y(x)=m*x+c
15
16 c=-2
17 m=1
18
19 fit y(x) "Lab-files/Data/2/Q_vi.txt" using 1:2 via m,c
20
21 plot "Lab-files/Data/2/Q_vi.txt" using 1:2 with p ps 1.5 lc "#009E73" title "Data
    points", \
22     y(x) lw 2 lc "#D55E00" title "Fitted line"
```

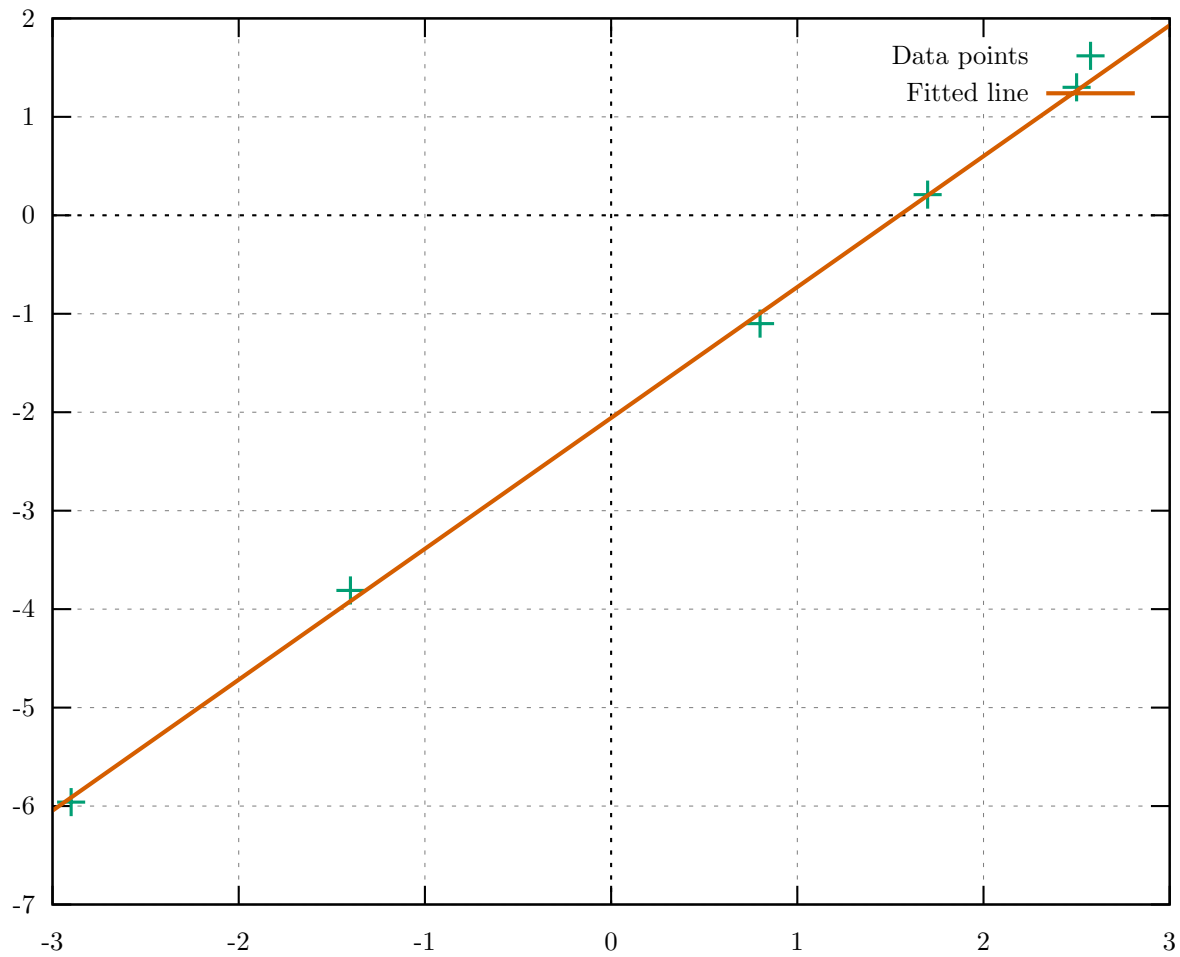


Figure 12: Linear fit for Problem Set 2 (vi).

Fit log (parameter estimates and standard errors):

Final set of parameters		Asymptotic Standard Error	
=====		=====	
m	= 1.32931	+/- 0.02098	(1.579%)
c	= -2.0581	+/- 0.04213	(2.047%)

Question 6

Problem Set 2 (vii). Fit the data with $f(x) = ax^2 + b$ over $-3 \leq x \leq 3$ and $0 \leq y \leq 16$. Plot the points and fitted curve, use the title *Polynomial fit*, and report a and b .

Table 6: Data for Problem Set 2 (vii).

x	y
-2.8	14.65
-1.2	5.81
0.7	4.48
1.8	8.33
2.9	15.56

Data file contents:

```
-2.8 14.65
-1.2 5.81
0.7 4.48
1.8 8.33
2.9 15.56
```

Sol: The fitted plot and the parameter estimates with their standard errors are shown below.

```
1  reset
2  set terminal svg background "white"
3  set ou "Lab-files/Output/2/Q_2_vii_ou.svg"
4
5  set fit logfile "Lab-files/GP-Files/2/Fit-log/fit_vii.log"
6  set title "Polynomial fit"
7  set grid
8  set xzeroaxis linestyle 0 lc "black"
9  set yzeroaxis linestyle 0 lc "black"
10
11 set xrange [-3:3]
12 set yrange [0:16]
13
14 y(x)=a*x**2+b
15
16 a=1
17 b=4
18
19 fit y(x) "Lab-files/Data/2/Q_vii.txt" using 1:2 via a,b
20 plot "Lab-files/Data/2/Q_vii.txt" using 1:2 with p ps 1.5 lc "#CC79A7" title "Data
    points", \
21     y(x) lw 2 lc "#0072B2" title "Fitted curve"
```

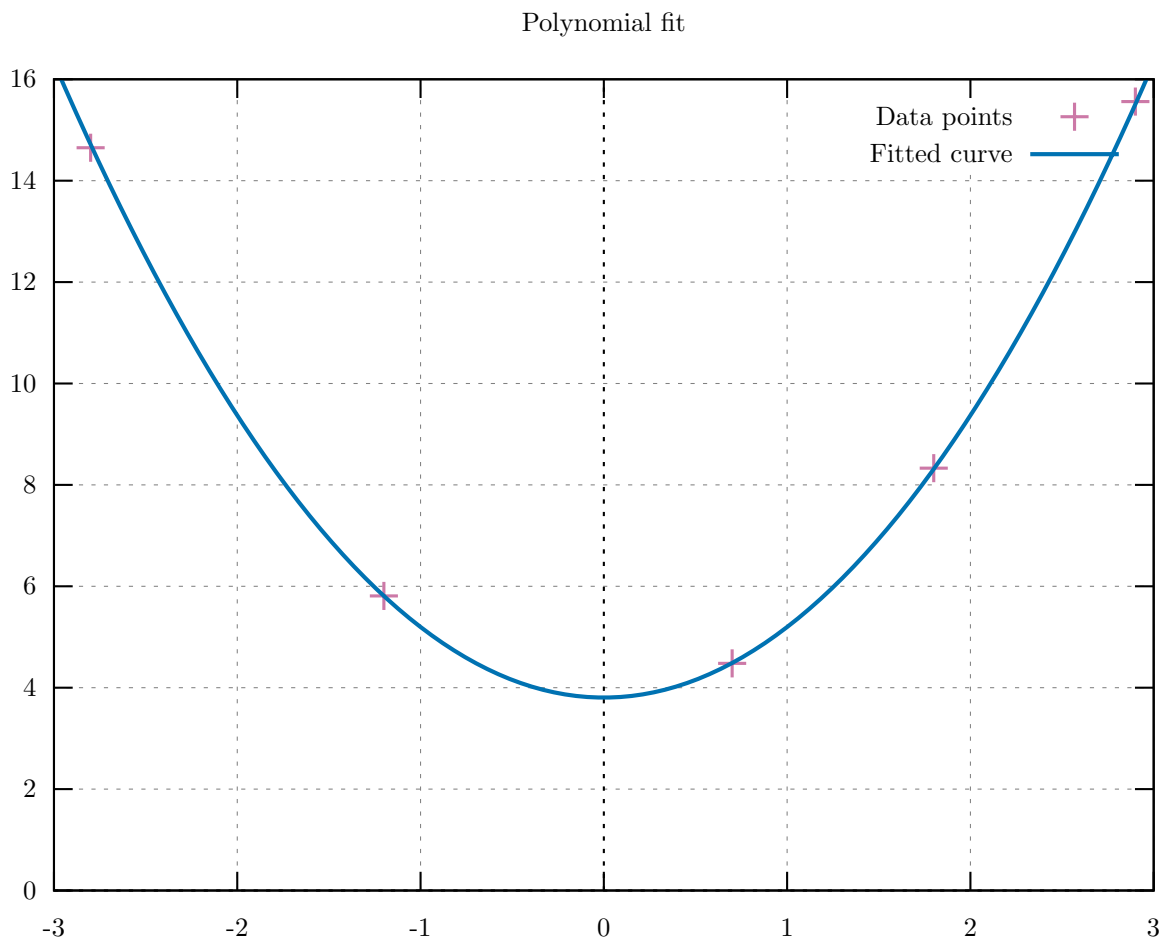


Figure 13: Polynomial fit for Problem Set 2 (vii).

Fit log (parameter estimates and standard errors):

Final set of parameters		Asymptotic Standard Error	
=====		=====	
a	= 1.39137	+/- 0.006722	(0.4831%)
b	= 3.80539	+/- 0.0362	(0.9513%)

Question 7

Problem Set 2 (viii). Fit the data with $f(x) = ae^{bx}$. Plot the points and fitted curve, use the title *Data fitting*, label the axes x and $f(x)$, and report a and b .

Table 7: Data for Problem Set 2 (viii).

x	$f(x)$
-3.0	3.422
-2.0	4.057
-1.0	4.808
0.0	5.7
1.0	6.756
2.0	8.008
3.0	9.492
4.0	11.251

Data file contents:

```
-3.0 3.422
-2.0 4.057
-1.0 4.808
0.0 5.7
1.0 6.756
2.0 8.008
3.0 9.492
4.0 11.251
```

Sol: The fitted plot and the parameter estimates with their standard errors are shown below.

```
1  reset
2  set terminal svg background "white"
3  set ou "Lab-files/Output/2/Q_2_viii_ou.svg"
4
5  set fit logfile "Lab-files/GP-Files/2/Fit-log/fit_viii.log"
6
7  set title "Data fitting"
8
9  set grid
10 set xzeroaxis linestyle 0 lc "black"
11 set yzeroaxis linestyle 0 lc "black"
12 set xlabel "x"
13 set ylabel "f(x)"
14
15 y(x)=a*exp(b*x)
16
17 a=1
18 b=1
19
20 fit y(x) "Lab-files/Data/2/Q_viii.txt" using 1:2 via a,b
21
22 plot "Lab-files/Data/2/Q_viii.txt" using 1:2 with p ps 1.5 lc "#0072B2" title "
    Data points", \
23     y(x) lw 2 lc "#D55E00" title "Fitted curve"
```

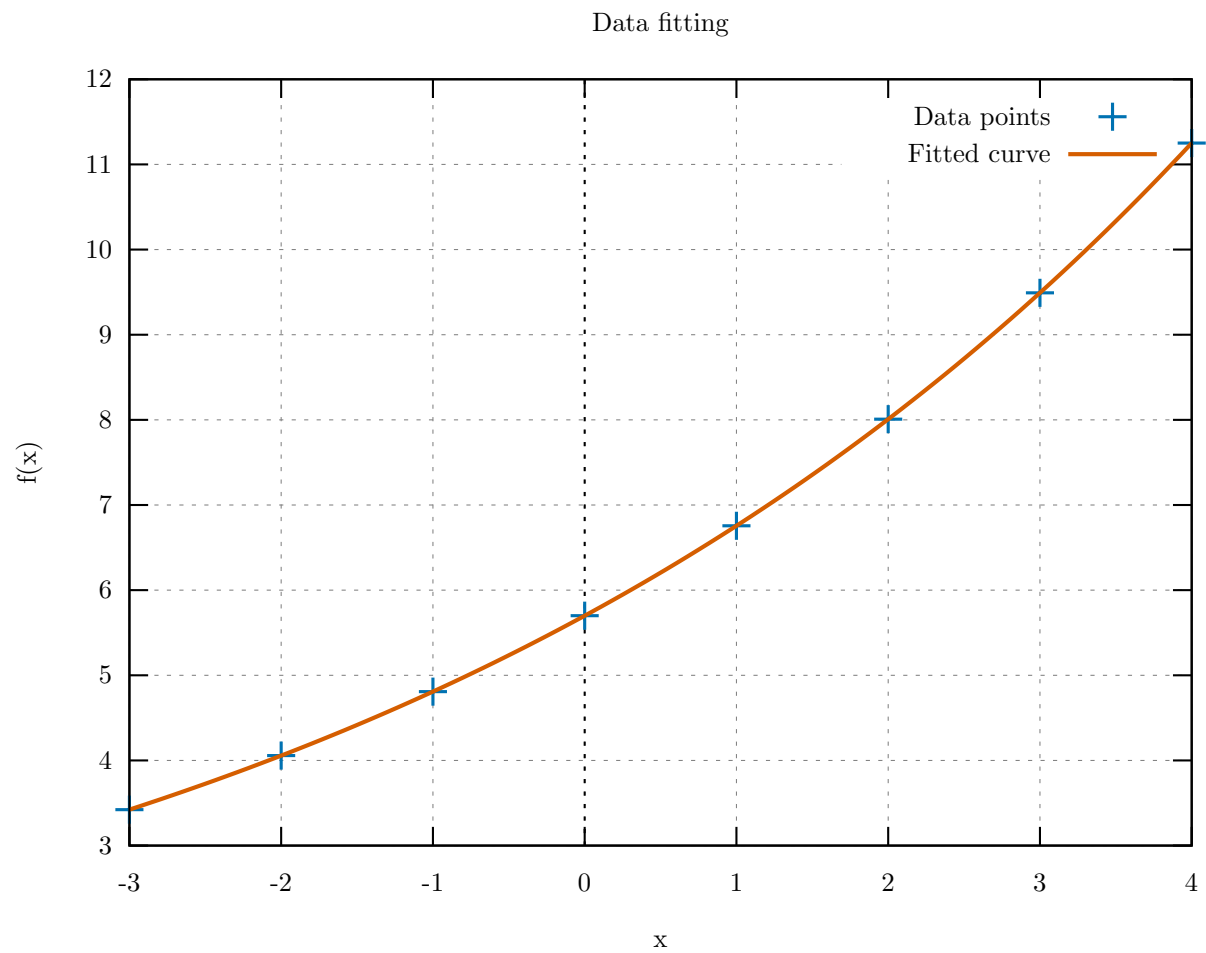


Figure 14: Exponential fit for Problem Set 2 (viii).

Fit log (parameter estimates and standard errors):

Final set of parameters		Asymptotic Standard Error	
=====		=====	
a	= 5.6996	+/- 0.0001403	(0.002462%)
b	= 0.170018	+/- 8.634e-06	(0.005078%)

Section 3

Question 1

Problem Set 3 (i). Show the polar plots of $\sin^3(t)$ and $\cos^2(2t)$. Label the axes “t value” and “the range”, place “polar plot” at $(0, -0.7)$, and show the vertical axis. **Sol:** The corresponding Gnuplot program and output are below.

```
1  reset
2
3  set polar
4  set terminal svg background "white"
5  set ou "Lab-files/Output/3/Q_3_i_ou.svg"
6
7  set title "Polar plot"
8  set grid polar linestyle 0 lc "black"
9  set xlabel "t value"
10 set ylabel "the range"
11 set label "polar plot" at 0,-0.7
12 set yzeroaxis linestyle 1 lc "black"
13
14 set samples 1000
15 plot (sin(t))**3 lw 2 lc "#0072B2",cos(2*t)**2 lw 2 lc "#D55E00"
```

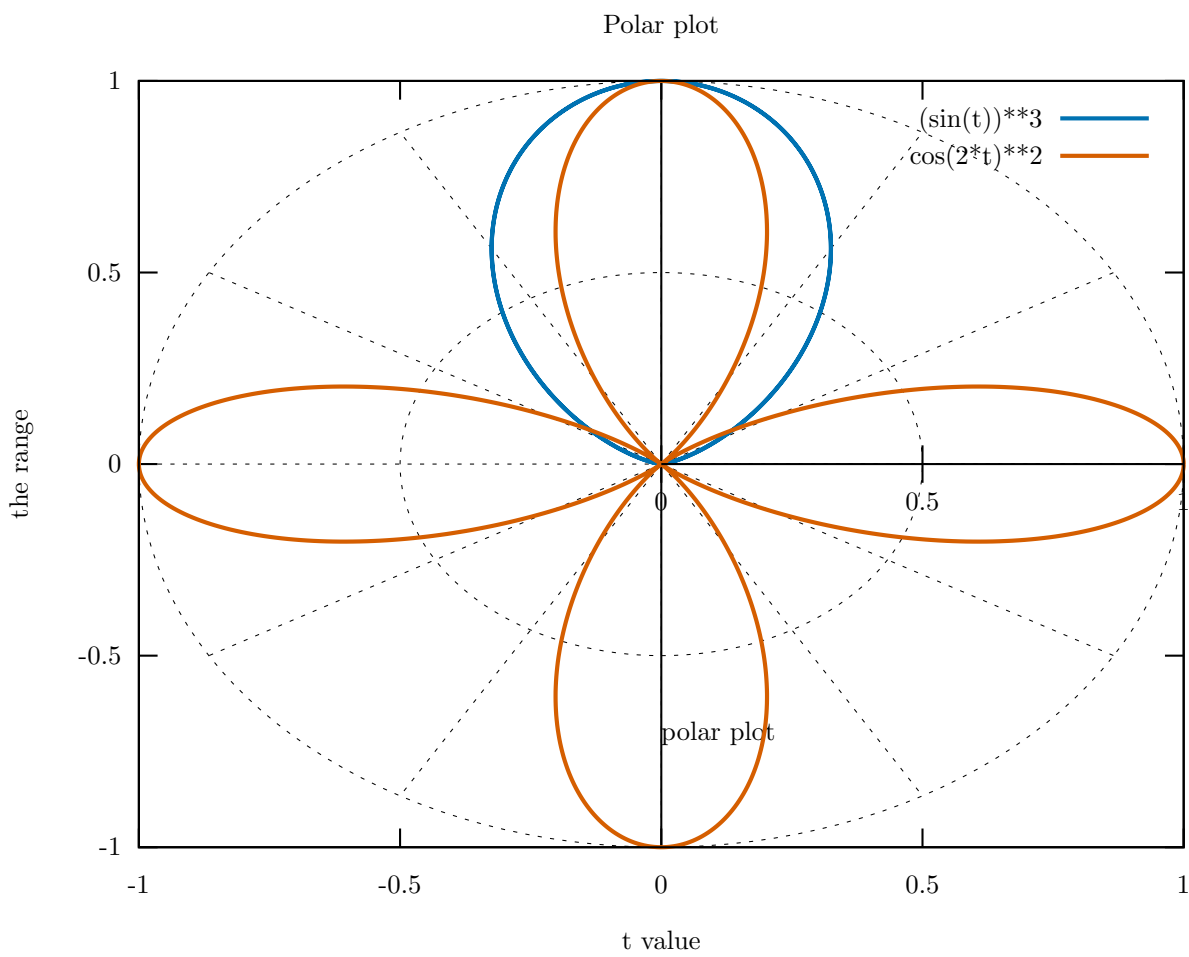


Figure 15: Polar plots of $\sin^3(t)$ and $\cos^2(2t)$.

Question 2

Problem Set 3 (ii). Show the polar plots of $\tan(6t)$ and $\cot(6t)$ over ranges -20 to 20 on both axes. Place the label “ $\tan(6t)$, $\cot(6t)$ ” at $(4, 12)$ and show both axes. **Sol:** The corresponding Gnuplot program and output are below.

```
1  reset
2
3  set polar
4  set terminal svg background "white"
5  set ou "Lab-files/Output/3/Q_3_ii_ou.svg"
6
7  set title "Polar plot"
8
9  set grid polar linestyle 1
10 set xlabel "t value"
11 set ylabel "the range"
12 set xzeroaxis linestyle 1 lc "black"
13 set yzeroaxis linestyle 1 lc "black"
14 set label "tan(6t), cot(6t)" at 4,12
15
16 set xrange [-20:20]
17 set yrange [-20:20]
18
19
20 set samples 1000
21
22 plot tan(6*t) lc "#0072B2", (1/tan(6*t)) lc "#D55E00"
```

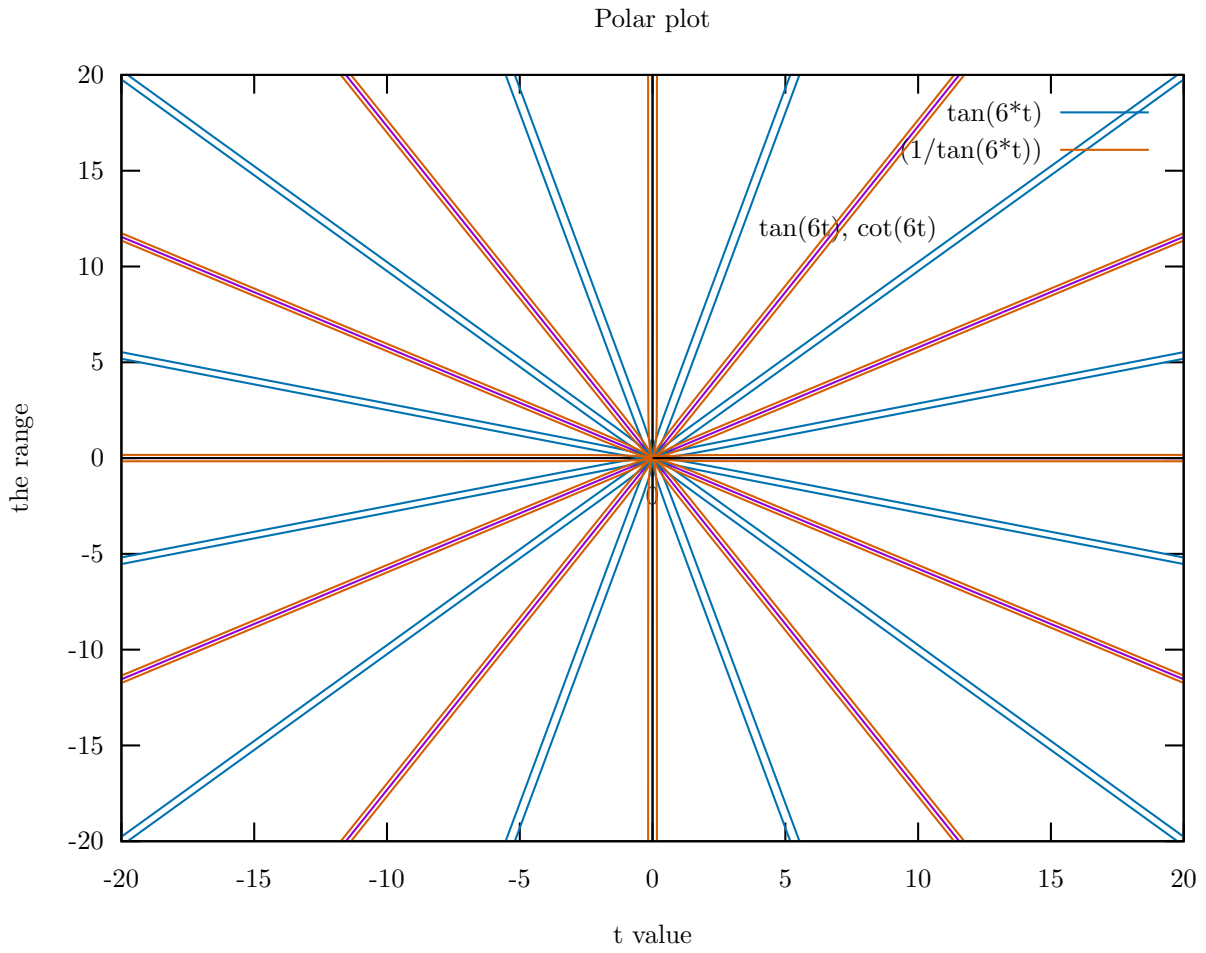


Figure 16: Polar plots of $\tan(6t)$ and $\cot(6t)$.

Question 3

Problem Set 3 (iii). Plot the parametric curve $x = 2 \cos^3 t$, $y = 2 \sin^3 t$ for $-\pi \leq t \leq \pi$. Label the axes as specified and title the plot “Astroid”. **Sol:** The corresponding Gnuplot program and output are below.

```
1  reset
2
3  set parametric
4  set terminal svg background "white"
5  set ou "Lab-files/Output/3/Q_3_iii_ou.svg"
6
7  set title "Astroid"
8  set xlabel "x = 2 cos(t)^3"
9  set ylabel "y = 2 sin(t)^3"
10 set grid linestyle 1 lc "black"
11 set xzeroaxis linestyle 1 lc "black"
12 set yzeroaxis linestyle 1 lc "black"
13
14
15 set trange [-pi:pi]
16 x(t)=2*cos(t)**3
17 y(t)=2*sin(t)**3
18
19 set samples 1000
20
21 plot x(t),y(t) lw 2 lc "#009E73"
```

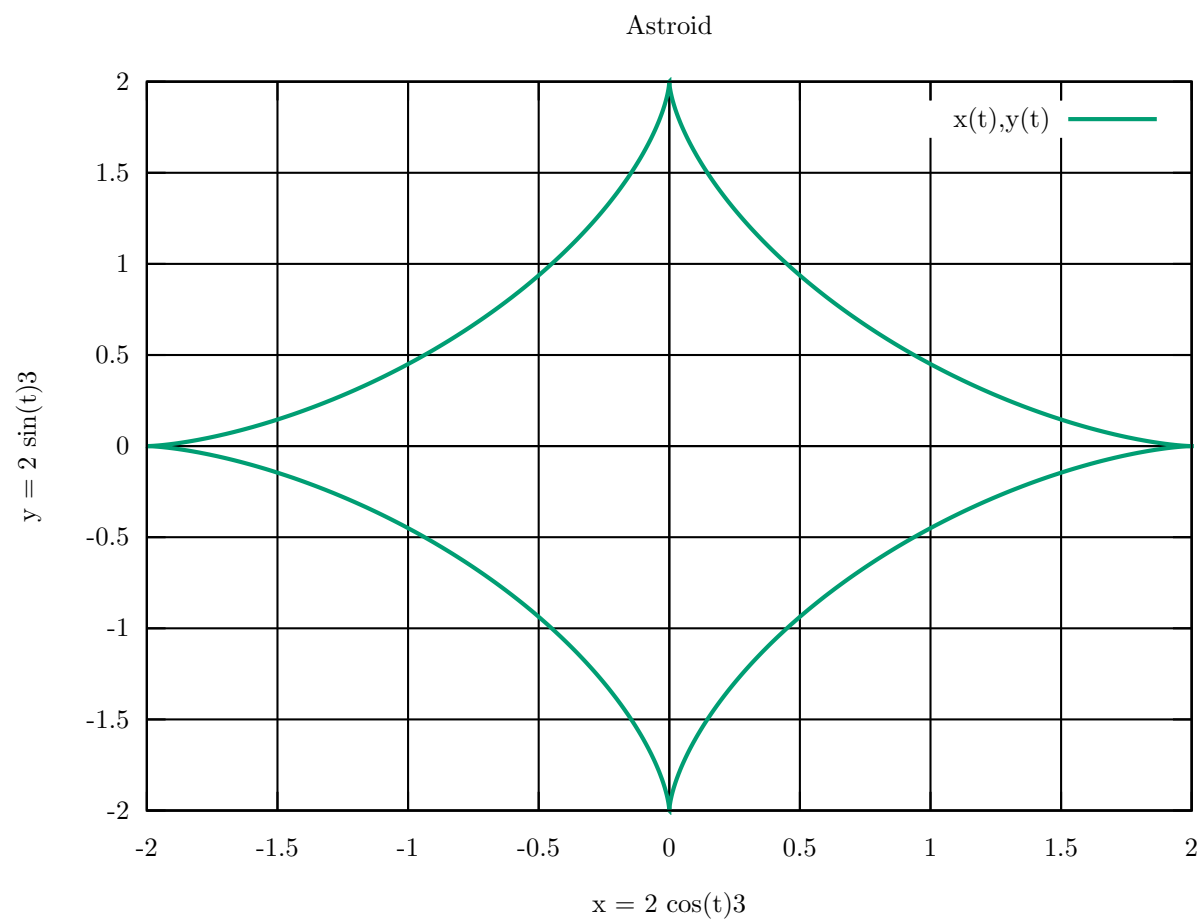


Figure 17: Astroid parametric plot.

Question 4

Problem Set 3 (iv). Plot the polar spiral $r = t$ for $0 \leq t \leq 6\pi$ using 10000 samples. Show both axes, label them as specified, and title the plot “Spiral”. **Sol:** The corresponding Gnuplot program and output are below.

```
1  reset
2
3  set polar
4  set terminal svg background "white"
5  set ou "Lab-files/Output/3/Q_3_iv_ou.svg"
6
7  set title "Spiral"
8  set xlabel "x = r cos t"
9  set ylabel "y = r sin t"
10 set grid polar lc "black"
11 set xzeroaxis linestyle 1 lc "black"
12 set yzeroaxis linestyle 1 lc "black"
13
14
15 set trange [0:6*pi]
16 r(t)=t
17
18 set samples 10000
19
20 plot r(t) lw 2 lc "#CC79A7"
```

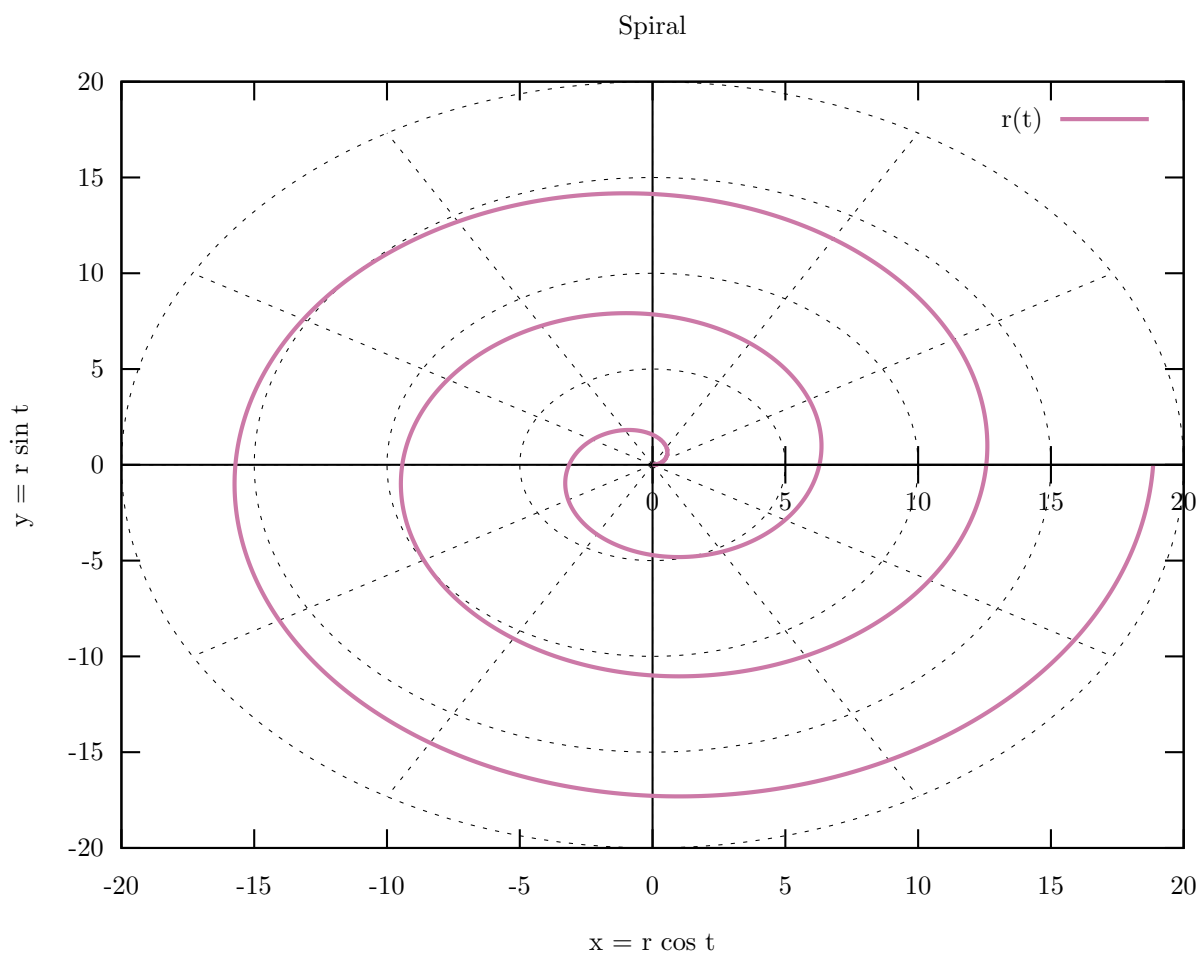


Figure 18: Polar spiral $r = t$.

Question 5

Problem Set 3 (v). Plot the parametric curve $x = 2 \sin t$, $y = 2 \sin(2t)$ for $-\pi \leq t \leq \pi$, with the specified axis labels and title “Lissajous Figure”. **Sol:** The corresponding Gnuplot program and output are below.

```
1  reset
2
3  set parametric
4  set terminal svg background "white"
5  set ou "Lab-files/Output/3/Q_3_v_ou.svg"
6
7  set title "Lissajous Figure"
8  set xlabel "x = 2 sin t"
9  set ylabel "y = 2 sin 2t"
10 set grid linestyle 0 lc "black"
11 set xzeroaxis linestyle 1 lc "black"
12 set yzeroaxis linestyle 1 lc "black"
13
14
15 set trange [-pi:pi]
16
17 x(t)=2*sin(t)
18 y(t)=2*sin(2*t)
19
20
21 set samples 1000
22
23 plot x(t),y(t) lw 2 lc "#D55E00"
```

Lissajous Figure

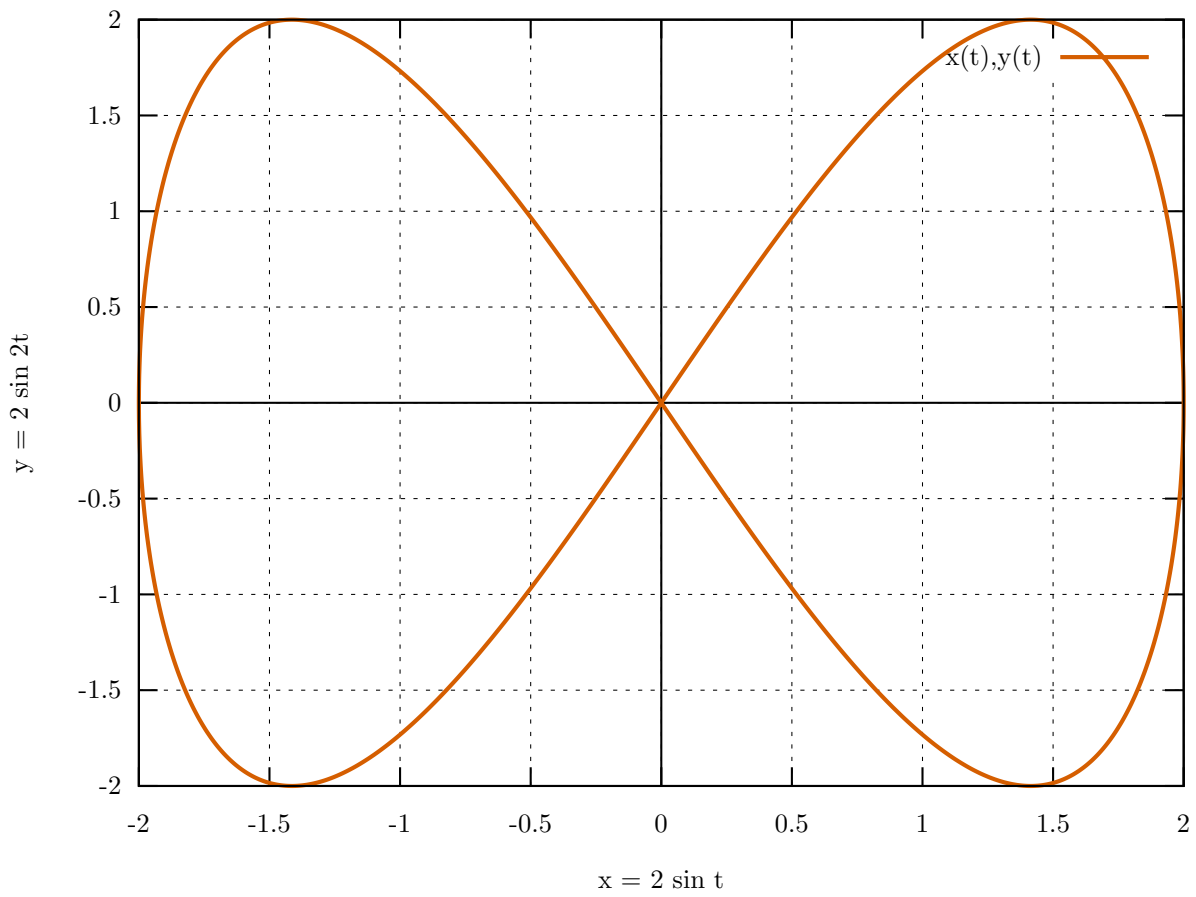


Figure 19: Lissajous parametric plot.

Question 6

Problem Set 3 (vi). Plot the polar curves $r_1 = 4 + 3 \cos t$ and $r_2 = 3 + 4 \cos t$. Label the axes as specified, place “polar plot” at $(-6, 6)$, and title the plot “A pair of Limacon”. **Sol:** The corresponding Gnuplot program and output are below.

```
1  reset
2
3  set polar
4  set terminal svg background "white"
5  set ou "Lab-files/Output/3/Q_3_vi_ou.svg"
6
7  set title "A pair of Limacon"
8  set xlabel "x = r cos t"
9  set ylabel "y = r sin t"
10 set grid linestyle 0 lc "black"
11 set xzeroaxis linestyle 1 lc "black"
12 set yzeroaxis linestyle 1 lc "black"
13 set label "polar plot" at -6,6
14
15 set trange [-pi:pi]
16
17 r1(t)=4+3*cos(t)
18 r2(t)=3+4*cos(t)
19
20
21 set samples 1000
22
23 plot r1(t) lw 2 lc "#0072B2", r2(t) lw 2 lc "#CC79A7"
```

A pair of Limacon

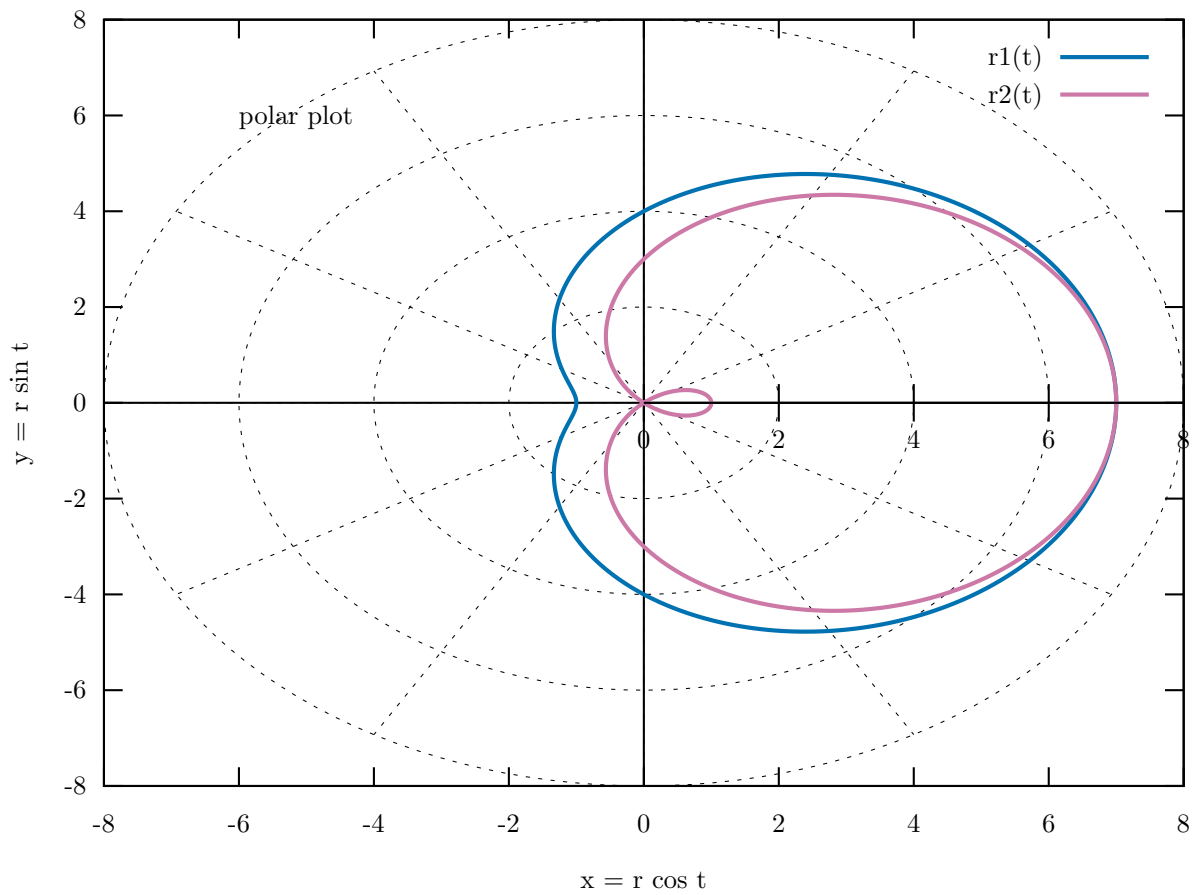


Figure 20: A pair of limaçons.